

MAKERERE UNIVERSITY
COLLEGE OF COMPUTING AND INFORMATION SCIENCES
SCHOOL OF COMPUTING & INFORMATICS TECHNOLOGY
END OF SEMESTER II EXAMINATION 2020/2021

PROGRAMME: BCSC (YR I)

YEAR OF STUDY: I

COURSE NAME: PROBABILITY AND STATISTICS

COURSE CODE: CSC 1201

DATE: Wednesday 17th August 2022

TIME: 8:00AM-11:00AM

EXAMINATION INSTRUCTIONS

- 1. ATTEMPT ALL QUESTIONS IN SECTION A (40 MARKS)**
- 2. ATTEMPT THREE (03) QUESTIONS IN SECTION B (60 MARKS)**
- 3. DO NOT OPEN THIS EXAM UNTIL YOU ARE TOLD TO DO SO**
- 4. ALL ROUGH WORK SHOULD BE IN YOUR ANSWER BOOKLET**
- 5. SHOW ALL THE NECESSARY WORKINGS**

SECTION A: (40 marks)

- 1) Explain the difference between a continuous random variable and discrete random variable. (4 marks)
- 2) Indicate which of the following variables are classified as discrete or continuous. (3 marks)
 - i) Number of typographical errors in newspapers
 - ii) **Length of a frog's jump**
 - iii) Number of cars owned by families
- 3) The time until a chemical is complete (in milliseconds) is approximated by the cumulative distribution function

$$F(x) = \begin{cases} 0, & x < 0 \\ 1 - e^{-0.01x}, & x \geq 0 \end{cases}$$

Determine the probability density function of X . What proportion of reactions is complete within 200 milliseconds? (5 marks)

- 4) Use the probability distribution given to answer the questions below and Let X represent number of calls.

x	0	1	2	3	4	5	6
$P(X=x)$	0.05	0.20	0.25	0.20	0.10	0.15	0.05

- i) Verify that the function is a probability mass function. (3 marks)
 - ii) Find the probability that there will be 3 or more calls. (3 marks)
 - iii) Find the probability that there will be at most 3 calls. (3 marks)
- 5) Write down the conditions under which the binomial distribution maybe a suitable model to use in statistical work. (3 marks)
- 6) Under what circumstance would one need to use the student's t -distribution when constructing confidence interval for the population mean? (2 marks)
- 7) Given $P(A) = 0.4$ and $P(A \cup B) = 0.6$. For what value of $P(B)$ are A and B independent? (4 marks)
- 8) Differentiate between descriptive statistics and inferential statistics (4 marks)
- 9) The following are the ages of six employees of an insurance company:

46 36 48 51 49 52

- i) Calculate the sample mean and sample variance for the above data. (4 marks)
 - ii) Of what importance is the mean and variance in summarizing a dataset. (2marks)

SECTION B: (60 MARKS)

QUESTION 1:

State the three axioms of probability.

(3 marks)

The two-way classification of married or previously married adults under 40 according to gender and age at first marriage produced the following table where E represents “people who were teenagers at first marriage”, W represents “people who were in their twenties at first marriage” and H represents “people who were in their thirties at first marriage”.

	E	W	H	TOTAL
M (male)	43	293	114	450
F (Female)	82	299	71	452
TOTAL	125	592	185	902

Determine whether or not the events F: “Female” and E: “was a teenager at first marriage” are independent.

(5 marks)

c) Two fair dice are rolled, let the scores obtained be X_1 and X_2 , and let $S = X_1 + X_2$ denote the total score.

i) What is the probability that both dice show an odd score? (2 marks)

ii) Calculate the expected value of S , $E(S)$ and the Variance of S , $var(S)$.

(4 marks)

iii) Calculate $P(S = 6 | X_1 \leq 3)$.

(6 marks)

QUESTION 2:

a) State the Central Limit theorem as used in statistics

(3 marks)

b) Consider a random sample $X_1, \dots, X_n$ from a distribution with unknown mean μ and known standard deviation σ . Assume that the sample size is large. Show that a $100(1 - \alpha)\%$ confidence interval for the mean μ is approximately;

$$\bar{X} \pm z_{(\alpha/2)} \frac{\sigma}{\sqrt{n}}$$

(7 marks)

c) The standard deviation for a population is $\sigma = 20$. A sample of size 100 observations selected from this population gave a sample mean equal to 62.

i) Construct a 90% confidence interval for the population mean μ . (5 marks)

ii) Provide an interpretation for your answer in i) above. (3 marks)

iii) Using examples from part c) above, differentiate between a parameter and a statistic.

(2 marks)

QUESTION 3:

- a) Let X denote the time in hours to locate and correct a problem in the software that governs the timing of traffic lights in the downtown area of Kampala. Assume X is normally distributed with a mean of 20 hours and variance of 16 hours. Find the probability that the next problem will require at least 24 hours to find and correct. (5 marks)

- b) Consider a continuous random variable X with probability density function

$$f_X(x) = \begin{cases} ax + bx^2, & \text{If } 0 < x < 1 \\ 0, & \text{Otherwise} \end{cases}$$

where a and b are constants so that this f_X is a valid density function and so that the expectation, $E(X) = 1/2$.

- Determine a and b . (7 marks)
- Give the Cumulative distribution function F_X of X . (4 marks)
- Calculate the $P(1/2 < X < 2)$. (4 marks)

QUESTION 4:

- a) The following data represent the lifetimes (in hours) of a sample of 25 transistors:

160	160	161	161	162	162	162	163	163	163
164	165	167	167	168	169	170	172	173	176
178	179	180	185	196					

- Construct a stem-and-leaf diagram for this dataset. Does the data appear skewed? If so, in which direction? (6 marks)
- Construct a boxplot for this data, clearly showing the median, lower and upper quartile as well as the minimum and maximum values of the dataset. (5 marks)
- Use the boxplot in ii) above to find the interquartile range. (2 marks)

- b) Sixty percent of new drivers in a particular country have had additional driving education. During their first year of driving, new drivers who have not had additional driving education have a probability of 0.09 of having an accident, while new drivers who have had additional driving education have a probability of 0.05 of having an accident.

- Calculate the probability that a new driver has had additional driving education, given that the driver had no accidents in the first year. (7 marks)

QUESTION 5:

Using examples, explain the difference between binomial distribution and hypergeometric distribution. (4 marks)

In clinical trials a certain drug has a 8% success rate of curing a known disease. If 15 people are known to have the disease.

- What is the probability that at least 2 are cured? (5 marks)
- Calculate the mean and variance of the number of people cured. (4 marks)

In a survey conducted by Makerere University Sports Club, respondents were asked "What is your favorite sport to watch?" Football and basketball ranked number 1 and 2 in terms of preference. In a group of 10 individuals, seven preferred football and three preferred basketball. If a random sample of three of these individuals is selected without replacement.

- With reasons, classify the kind of distribution in scenario c) above. (2 marks)
- What is the probability that exactly two prefer football? (5 marks)

“END”

Appendix 1: Sample Formulas

$$P_x = \binom{n}{x} p^x q^{n-x}$$

$$p_X(k) = \Pr(X = k) = \frac{\binom{K}{k} \binom{N-K}{n-k}}{\binom{N}{n}}$$

$${}_nC_r = \frac{n!}{r!(n-r)!}$$